

SYED AFFAN ALI

Computer Systems Engineering | Robotics, IoT & Applied AI
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RESEARCH INTERESTS

AI-Driven Robotics & Autonomous Systems; Computer Vision; Multimodal AI & Human-Robot Interaction; IoT & Intelligent Sensing Systems; Edge AI & Embedded Machine Learning (TinyML); Agentic AI;

RESEARCH OBJECTIVE

To pursue a Master's degree in Artificial Intelligence, Robotics, or Intelligent Systems, building on hands-on experience in embedded hardware, sensor integration, and connected systems. My goal is to develop research skills in computer vision, autonomous robotics, and edge AI and contribute to intelligent systems through prototyping, experimentation, and analysis.

EDUCATION

Bachelor of Engineering — Computer Systems
Mehran University of Engineering & Technology (MUET) | Jamshoro, Pakistan | 2019 – 2023
CGPA: 2.88 (3.34 GPA in Final Year)

PROJECTS

Gesture-Controlled Smart Robotic Wheelchair

ESP | Arduino | Gyroscope | Sonar | Proximity & Touch Sensors | GSM | Motor Control

- Engineered an assistive mobility prototype controlled through hand gestures using ESP and Arduino boards with responsive motor control.
- Integrated gyroscope, sonar, proximity, and touch sensors with GSM connectivity and battery-management circuits for connected operation and safety monitoring.

2D Ultrasonic Radar Mapping Prototype (Poor Man's LiDAR)

Arduino | HC-SR04 | Stepper Motor / Servo | Python | Matplotlib | USB Serial

- Built a low-cost 2D mapping prototype that rotates an HC-SR04 ultrasonic sensor with a stepper motor or servo to scan the surrounding environment.
- Streamed angle-and-distance readings from Arduino over USB serial and rendered them on a live Python polar plot using Matplotlib.

IoT-Based Water Quality Monitoring System

Arduino | Python | pH, Turbidity & Temperature Sensors | IoT | Data Visualization

- Designed a real-time water-quality monitoring system using Arduino with pH, turbidity, and temperature sensors.
- Built a Python pipeline for normalization, remote visualization, anomaly detection, and threshold-based alerts, with wireless data transmission for remote monitoring.

AI-Based IoT Energy Monitoring & Anomaly Detection

ESP32 | Python | Machine Learning | IoT | Sensors

- Built an ESP32 prototype that streams electrical measurements for remote monitoring and analysis.
- Applied a lightweight anomaly-detection model to flag unusual energy-consumption patterns and present results through a monitoring interface.

Smart Greenhouse Monitoring & Prediction System

ESP32 | Python | Temperature, Humidity & Light Sensors | Machine Learning | IoT

- Collected temperature, humidity, and light readings with an ESP32 and made the environmental data available for remote monitoring.
- Applied a simple machine-learning model for near-term condition prediction and actionable threshold alerts.

AI-Powered Document Q&A Agent

Python | LangChain | OpenAI API | FAISS | RAG | Streamlit

- Built a retrieval-augmented document assistant that ingests PDF and text files, performs FAISS vector search, and answers questions with source citations.
- Implemented conversational memory and a Streamlit upload-and-query interface for coherent multi-turn document exploration.

WORK EXPERIENCE

Mobile Application Developer

Gentec Pvt. Ltd. | Karachi, Pakistan | 11/2024 – Present

- Develop Flutter/Dart feature modules with clean architecture and scalable state management, contributing to code reviews, sprint delivery, and CI/CD workflows.

Mobile Developer · IoT & Connected Apps

Wamtsol Pvt. Ltd. | Karachi, Pakistan | 02/2024 – 10/2024

- Built cross-platform Flutter apps with Provider/Bloc, native device APIs, REST integration, and offline persistence and background uploads using IsarDB, SharedPreferences, and Dart isolates.

Flutter Front-End Developer

BaijiSoft Technologies | Karachi, Pakistan | 08/2023 – 01/2024

- Translated Figma designs into responsive Flutter interfaces and integrated REST APIs, JSON data flows, and multipart requests across devices.

Mobile Application Developer Intern

Zensoft Pvt. Ltd. | Karachi, Pakistan | 06/2023 – 07/2023

- Built a field-deployed Sugar Cane Trolley Registration app with geolocation, image capture, Google Maps, and offline background uploads using Dart isolates.

TECHNICAL SKILLS

Embedded & Robotics: Arduino | ESP32 | C / C++ | Microcontroller Interfacing | Sensor Integration | Wireless Telemetry | Gesture-Controlled Robotics | IoT

AI / ML: Python | TensorFlow | PyTorch | Scikit-learn | Hugging Face Transformers | LangChain | RAG Pipelines | Prompt Engineering

Computer Vision: OpenCV | Image Processing | Object Detection | Gesture Recognition | Real-time Video Analysis

Agentic AI: LangChain Agents | Tool Use | LLM Orchestration | OpenAI API | Autonomous Pipelines

Programming Languages: Dart | Python | C | C++ | JavaScript (basic)

Mobile Development: Flutter | Dart | Provider | Bloc | IsarDB | SharedPreferences | Dart Isolates | REST APIs | JSON | HTTP/Multipart

Tools & Practices: Git | GitHub | Figma | VS Code | Agile/Scrum | Unit Testing | SOLID Principles | Clean Architecture

CERTIFICATIONS & TRAINING

Mobile App Development with Flutter | MUET | 07/2022 – 12/2022

Arduino & Robotics | Creativo, Karachi | 04/2021 – 06/2021

IELTS English Proficiency | Band Score: 6.5